The Game Of Un-Impartial Kayles *

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Abstract

This article introduces an un-impartial version of an old English game named Kayles. We show how combinatorial game theory can be applied to solve the game. Un-impartial Kayles has periodic position values that are sums of $\uparrow$ *and* $*$.

1 Introduction

Combinatorial game theory[1][2] has been the basic common mathematical model for analyzing intelligent games since 1970s. It starts from by defining a game simply, i.e. , as an ordered pair of sets of games. Conventionally, game G is denoted as:

$$G = \{G^L | G^R\}, \tag{1}$$

where G^L and G^R are sets of games.

When both G^L and G^R are empty sets ($\emptyset$), the game is special, and named 0.

$$0 = \{\emptyset|\emptyset\}. \tag{2}$$

Each game has a negation; the negation of game G is defined as

$$-G = \{-G^R| - G^L\}. \tag{3}$$

The sum of games G and H is defined as

$$G+H = \{G^L+H, G+H^L|G^R+H, G+H^R\}. \tag{4}$$

A partial order is defined on the sets of games:

$$G \geq 0 \text{ if and only if no element exists in } G^R \leq 0, \tag{5}$$

$$G \leq 0 \text{ if and only if } -G \geq 0, \tag{6}$$

$$G \geq H \text{ if and only if } G - H \geq 0. \tag{7}$$

*This paper is supported by the National Science Council, Taiwan, R.O.C. (NSC-96-2221-E-346-004).

An equivalence relation may be further defined on the sets of games:

$$G \equiv H \text{ if and only if } G \geq H \text{ and } G \leq H. \tag{8}$$

Equivalence classes form an algebraic group.

2 Kayles

The game of *Kayles* was introduced by Dudency and also by Sam Lord[1]. Two players, L and R, take turns knocking down one single pin or two adjacent pins in a row of bowling pins. The player who knocks down the last pin is the winner. Kayles is a classical *impartial* game[4]; each position of Kayles is equivalent to some *nimber*[3]. The nimbers for Kayles also exhibit periodicity. The period is 12, but with 14 exceptional values at 0, 3, 6, 9, 11, 15, 18, 21, 22, 28, 34, 39, 57 and 70.

n	0	12	24	36	48	70	82
0	*0	*4	*4	*4	*4	*4	*4
1	*1	*1	*1	*1	*1	*1	*1
2	*2	*2	*2	*2	*2	*2	*2
3	*3	*7	*8	*3	*8	*8	*8
4	*1	*1	*5	*1	*1	*1	*1
5	*4	*4	*4	*4	*4	*4	*4
6	*3	*3	*7	*7	*7	*7	*7
7	*2	*2	*2	*2	*2	*2	*2
8	*1	*1	*1	*1	*1	*1	*1
9	*4	*4	*8	*8	*4	*8	*8
10	*2	*6	*6	*2	*2	*2	*2
11	*6	*7	*7	*7	*7	*7	*7

Table 1. Periodicity of Kayles nimbers.

In un-impartial Kayles, L is allowed to knock down one single ping at a time. R is allowed to knock down two adjacent pins at a time, except for a single-pin row. In the later case, R is allowed to knock down one single pin. Un-impartial Kayles envolves infinitesimals other than nimbers, which are reviewed in the next section.

3 Up and Down

An *infinitesimal* is a game less than any positive numbers and greater than any negative numbers. All the nimbers are impartial infinitesimals. The simplest un-impartial infinitesimal is $\uparrow$ (pronounced as *up*). It is defined as

$$\uparrow = \{0|*\} \tag{9}$$

where

$$* = \{0|0\} \tag{10}$$

(pronounced as *star*). Note that

$$* + * = 0. \tag{11}$$

The negation of $\uparrow$ is denoted as $\downarrow$ (pronounced as *down*). $\uparrow$ is greater than 0, but confused with $*$. With the help of the inequality:

$$\uparrow + \uparrow > *, \tag{12}$$

the outcome of any sum of $\uparrow$, $\downarrow$ and $*$ can be determined [1].

Outcome Rule

A sum G of $\uparrow$, $\downarrow$ and $*$ is 0 if and only if there is an even number of $*$ and the number of $\uparrow$ equals the number of $\downarrow$, and, *positive* ($G > 0$) if and only if

1. there is an even number of $*$ and the number of $\uparrow$ is more than the number of $\downarrow$, or
2. there is an odd number of $*$ and the number of $\uparrow$ is at least two more than the number of $\downarrow$.

Once one knows $G = 0$, $G > 0$, $G < 0$ or none of the above, the outcome of G can be determined by Table 2.

$G = 0$,	the second player wins.
$G > 0$,	L wins.
$G < 0$,	R wins.
Otherwise,	the first player wins.

Table 2. Outcome of a game.

In the next section, we shall see that any position in un-impartial Kayles is simply is a sum of $\uparrow$, $\downarrow$, and $*$, and therefore the outcome can be determined by the rule above.

[1] Conway[2] introduced a more extended family of ups. He provided a simple rule (Theorem 88 in [2]) to determine the outcome of any sum of these extendend ups, downs and star.

4 Un-Impartial Kayles

We use the notation $U(n)$ to denote the G value of a row with n pins in un-impartial Kayles. When $n = 0$, none of the players has a legal move.

$$U(0) = \{\ |\ \} = 0. \tag{13}$$

When $n = 1$, both of the players have the same option to knock down a pin.

$$U(1) = \{0 \mid 0\} = *. \tag{14}$$

A row with $n \geq 2$ pins has a G value of the form:

$$U(n) = \{U(a) + U(b) : a + b = n - 1 |$$
$$U(c) + U(d) : c + d = n - 2\}$$

where $a, b, c, d \geq 0$.
When $n = 2$,

$$U(2) = \{U(1) \mid 0\} = \{* \mid 0\} = \downarrow. \tag{15}$$

When $n = 3$,

$$U(3) = \{U(2), U(1) + U(1) \mid U(1)\}$$
$$= \{\downarrow, 0 \mid *\} = \{0 \mid *\} = \uparrow. \tag{16}$$

Continue the analysis up to $n = 7$, we have [2]

$$U(4) = \{\uparrow, \downarrow * \mid \downarrow, 0\} = \{\uparrow \mid \downarrow\} = *, \tag{17}$$

$$U(5) = \{*, \uparrow *, \downarrow\downarrow \mid \uparrow, \downarrow *\} = \{\uparrow * \mid \downarrow *\} = 0, \tag{18}$$

$$U(6) = \{0, 0, 0 \mid *, \uparrow *, \downarrow\downarrow\} = \{0 \mid \downarrow\downarrow\} = \downarrow *, \tag{19}$$

$$U(7) = \{\downarrow *, *, \downarrow *, \uparrow\uparrow \mid 0, 0, 0\} = \{\uparrow\uparrow \mid 0\} = \uparrow *. \tag{20}$$

When $n \geq 8$, the game has a period of 8. The proofs are provided in the next section.

n	U(n)
8m+0	0
8m+1	$*$
8m+2	$\downarrow$
8m+3	$\uparrow$
8m+4	$*$
8m+5	0
8m+6	$\downarrow *$
8m+7	$\uparrow *$

Table 3. Periodicity of un-impartial Kayles.

[2] In general, to prove $A = B$ where A and B are games, it is sufficient to show that the first player cannot win the game $A - B$.

5 Proof of Periodicity

Proposition:

$$U(8m+k) = U(k), m \geq 0, 0 \leq k < 8. \quad (21)$$

Proof: We prove by induction on m. When $m = 0$, the claim is trival. Suppose $U(8m+k) = U(k), m \geq 0, 0 \leq k < 8$, we want to show $U(8(m+1)+k) = U(k), m \geq 0, 0 \leq k < 8$.
When $k = 0$,

$$\begin{gathered} U(8(m+1)+k) = U(8m+8) \\ = \{U(a)+U(b) : a+b = 8m+7 | \\ U(c)+U(d) : c+d = 8m+6\} \\ (apply\ induction\ hypothesis) \\ = \{U(a)+U(b) : a+b = 7 | \\ U(c)+U(d) : c+d = 6 \text{ or } c = 7, d = 7\} \\ = \{\uparrow *, \downarrow, \downarrow, \uparrow * | \downarrow *, *, \downarrow *, \uparrow\uparrow, \uparrow\uparrow\} \\ (eliminate\ dominated\ options) \\ = \{\uparrow * | \downarrow *\} \\ (apply\ simplification\ rule) \\ = 0 \end{gathered}$$

When $k = 1$,

$$\begin{gathered} U(8(m+1)+k) = U(8m+9) \\ = \{U(a)+U(b) : a+b = 8m+8 | \\ U(c)+U(d) : c+d = 8m+7\} \\ = \{U(a)+U(b) : a+b = 8, a \geq 1 | \\ U(c)+U(d) : c+d = 7\} \\ = \{\uparrow, \downarrow\downarrow, \uparrow, 0 | \uparrow *, \downarrow, \downarrow, \uparrow *\} \\ = \{\uparrow | \downarrow\} \\ = * \end{gathered}$$

When $k = 2$,

$$\begin{gathered} U(8(m+1)+k) = U(8m+10) \\ = \{U(a)+U(b) : a+b = 8m+9 | \\ U(c)+U(d) : c+d = 8m+8\} \\ = \{U(a)+U(b) : a+b = 9, a \geq 2 | \\ U(c)+U(d) : c+d = 8, c \geq 1\} \\ = \{\uparrow *, \downarrow, \downarrow, \uparrow * | \uparrow, \downarrow\downarrow *, \uparrow, *\} \\ = \{* | \downarrow\downarrow *\} \\ = \downarrow \end{gathered}$$

The proofs for $3 \leq k \leq 7$ are similar to the ones above. □

6 Example

This finnal section provides a complete example demostrating how to analyze a game of un-impartial Kayles. Consider an un-impartial Kayles with 3 rows of pins. The number of pins in the rows are 3, 5, and 12. Who will win the game? To analyze a game, one first checks Table 3 to determine the G-value of the game. In this example, the G-value is

$$G = U(3) + U(5) + U(12) = \uparrow + 0 + * = \uparrow *.$$

Next, apply the outcome rule described in section 2 to determine whether $G > 0$, $G < 0$, $G = 0$ or not. In this example, $G \not> 0$, $G \not< 0$ and $G \neq 0$. Finally, check Table 2 to detemine the winner. In this example, the player who moves first is the winner.

The game of un-impartial Kayles is simple and easy, from the prospect of either rule or analysis. Both Kayles and un-impartial Kayles exhibit periodicity. It is interesting to see that un-impartial Kayles has no exception value of its periodicity.

References

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[3] C. L Bouton , *Nim : A Game With a Complete Mathematical Theory*, Annals of Math, Vol. 3, pp. 35-39.

[4] P. M. Grundy, *Mathematics and Games*, Eureka, 1939.